

SHREE CHOUDHARI

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EDUCATION

SYMBIOSIS SKILLS & PROFESSIONAL UNIVERSITY

Bachelor of Technology in Mechatronics Engineering (CGPA-9.0057)

Pune, India

AUG 2026

Relevant Coursework: Automation, Robotics, Embedded Electronics, Application and Device Development.

GOVERNMENT POLYTECHNIC, NAGPUR

Diploma in Mechatronics Engineering (86%)

Nagpur, India

July 2023

Relevant Coursework: PLC, SCADA, Power Electronics, Drive Systems, Rapid prototyping.

SKILLS

TECHNICAL SKILLS: Rapid Prototyping, PLC Programming, 3D Modeling and Design, Circuit Design, Simulation and Analysis, Microcontroller Programming, CFD Analysis, SCADA System Design, PCB Design and Layout, PCB Fabrication, Industrial IoT Integration, Sensor Integration & Calibration, Laser Cutting, Arduino Programming.

Software Skills: C++, Python, Totally Integrated Automation (TIA Portal), RoboAnalyzer, InTouch (SCADA), KiCad EDA, Autodesk® Fusion 360, Keil, AutoCAD, Ansys computational fluid dynamics (CFD), SolidWorks, MATLAB, RUIDA RD Works, UltiMaker Cura, Fracktory Slicer, Proteus, Multisim, RD Works (CNC Laser Cutting). MS Office (Word, Excel, Power Point, Outlook)

INTERNSHIP & RESEARCH EXPERIENCE

TATA ADVANCED SYSTEMS LIMITED (TASL), Land Mobility Division, PUNE

Pune, India

Senior Project Intern

June 2025 – July 2026

Creation and Integration of AR/VR Modules to enhance Employee skills in Manufacturing Operations for India and Morocco

- **VR Training Module Development:** Developed Virtual Reality (VR) welding training modules integrating industrial-grade sensors with welding equipment, improving training quality and safety across Indian and Moroccan facilities.
- **Technical Leadership:** Led operator training, fostering teamwork and boosting workforce proficiency through innovative metric-based feedback systems.
- **System Optimization:** Collaborated with vendors to troubleshoot and optimize simulation systems for latency, interface design, and stability.
- **Cross-Cultural Communication:** Assisted in development of a offline, real-time multilingual translator to bridge communication gaps between Indian employees and foreign delegates.
- **AI automation:** Contributed to the development of enterprise digital solutions by exploring AI automation, Large Language Models (LLMs), and intelligent workflow applications for manufacturing operations.

TATA MOTORS PASSENGER VEHICLE LIMITED (TMPVL), PUNE

Pune, India

Project Intern

June 2024 – August 2024

New Model Introduction (Punch MY 24)

- **VQA Data & PPAP Management:** Managed Vendor Quality Assurance (VQA) data and oversaw Production Part Approval Process (PPAP) compliance.
- **SAP & IPMS Proficiency:** Utilized SAP for project data management and IPMS for real-time vehicle tracking.
- **Manufacturing & Quality Processes:** Gained expertise in vehicle manufacturing, performed pre-delivery quality checks, and assisted in Vehicle Line Optimization (VLO).
- **Proving Plan & Analysis:** Prepared proving plans, reviewed rollup for 35+ BIW variants, and conducted Track Configuration (TCF) validation.
- **Readiness Assessments:** Ensured Press/BIW, Facility, Software, Logistics, and VQA readiness, evaluating Manufacturing Readiness
- **Cross-Functional Collaboration:** Worked with TS, CPA, VQA, Craftsmanship, and Logistics teams to ensure seamless operations.
- **Additional Exposure:** Assisted in wiring harness integration, radiator testing, and gained insights into AGV maintenance & ASRS operations.

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY (IIT BOMBAY)

Mumbai, India

Senior Project Intern

April 2023 – August 2023

Development of a Medical Device for Sonophoresis-Based Transdermal Drug Delivery

- Designed and developed a medical device utilizing sonophoresis for enhanced transdermal drug delivery.
- Conducted experimental validation and optimized device parameters for efficient drug permeation.

Iontophoresis-Based Transdermal Drug Delivery System

- Worked on a project focused on iontophoresis technology to enhance drug penetration through the skin.
- Assisted in prototype development and performance evaluation.

Design & Rapid Prototyping of Microneedle Molds for Drug Delivery

- Designed and fabricated microneedle molds for transdermal drug delivery applications.
- Utilized rapid prototyping techniques to develop and refine mold structures for enhanced precision and functionality.

HALDIRAM FOODS INTERNATIONAL LTD, NAGPUR

Nagpur, India

Project Intern

February 2023 – March 2023

- **Designed an Automatic Pistachio Cleaning Machine** to improve efficiency and reduce manual labor.
- **Developed an AGV Layout** for factory automation and pitched the use of Automated Guided Vehicles (AGVs) to optimize manufacturing processes.

- **Developed an IoT-Based Emergency Fire Response System** for automatic fire exit unlocking and real-time alerts to the fire department.
- **Performed Maintenance & Troubleshooting** on existing machinery to ensure optimal performance and minimize downtime.

FABLAB

Nagpur, India

September 2023 – December 2023

- **PRECISION MANUFACTURING:** Gained experience with laser cutting machines, focusing on precision manufacturing of complex geometries.
- **WORKED WITH VARIOUS MATERIALS:** Such as different types of acrylics, and MDF boards for projects.
- **ADVANCE MANUFACTURING:** Created prototypes using FDM and SLA 3D printers, focusing on reducing lead time as well as material consumption.
- **USED FUSION 360:** To design 3D models and prepare them for various projects and clients, with minimal post-processing requirements.
- **MATERIAL EXPERTISE:** worked with materials like PLA, ABS, and resin to produce durable and lightweight components for mechanical systems

PATENTS PUBLISHED

1. MOBILITY AID WITH ADVANCED NAVIGATION AND EMERGENCY RESPONSE FOR THE VISUALLY IMPAIRED. (UTILITY PATENT)
2. INTEGRATED COMMUNICATION ENABLED NAVIGATIONAL SUPPORT DEVICE FOR THE VISUALLY IMPAIRED. (DESIGN PATENT) **Design no.:** 436426-001 | **Status:** Granted & Registered (21/04/2025)
3. **Copyright:** Smart Mobility Aid With AI-powered Navigation and Emergency Assistance for the Visually Impaired **Registration Number:** LD-20250169851 | **Status:** Registered (05/08/2025)
4. *Autonomous Modular Multi-Payload Aerial System with Adaptive AI Control Architecture* **CBR Number & Date:** 210968 | 09-05-2026 **Status:** Filed (Under Secrecy Direction)

ACADEMIC PROJECTS

Mobility Aid with Advanced Navigation & Emergency Response for the Visually Impaired

- Developed a smart Cane stick integrating Arduino & Raspberry Pi for real-time obstacle detection, fall detection, and emergency response.
- Engineered an ultrasonic sensor-based obstacle detection system providing haptic feedback for safer navigation. Implemented a GPS & GSM module (A9G) to enable live tracking and automatic distress signal transmission.
- Integrated piezoelectric sensors for fall detection, triggering emergency alerts and sending location & images via Raspberry Pi Camera. Designed a solar-powered energy-efficient system with hands-free communication for enhanced user safety.

Designed & Fabricated an FDM-Based 3D Printer

- Built a fully functional FDM 3D printer from scratch with LAN technology for improved efficiency and remote operation.
- Integrated filament-saving sensors to prevent material wastage during failed prints.

Customization and Integration of Assembly Line for Industrial Automation

- Using Siemens S7-300 PLC for process control and monitoring, an automated assembly line was developed and integrated.
- Programmed and configured the complete system in TIA Portal.
- Worked with various industrial sensors and actuators for real-time feedback and control.

Chemical Mixing Control Process through HMI

- Developed an automated chemical mixing process controlled via HMI integrated with a Siemens PLC system.
- Wrote the complete ladder logic program in TIA Portal and configured HMI tags for real-time process monitoring and control.
- Designed and customized the HMI screens according to process requirements for improved operator interface and usability.

Robotic Hand Controlled via Flex Sensors

- Built a robotic hand that mimics human movements using flex sensors, allowing for gesture-based control.

Developed a PET Filament Machine for Sustainable 3D Printing

- Engineered a system that converts plastic bottles into 3D printer filament, promoting eco-friendly additive manufacturing.
- Designed a heated nozzle system to process PET bottle ribbons into uniform filament for smooth extrusion.

EXTRA-CURRICULAR AND RESEARCH PAPERS

- Showcased a project at the State Level Project Competition (2023).
- Executive Member of Indian Society for Technical Education (2022)
- Organized Robotics Competition at University Level (2025)
- AN EXTENSIVE EXPLORATION IN INTEGRATION OF ROBOTICS TO AMPLIFY EFFICIENCY WITHIN LOGISTICS THROUGH AUTOMATED STORAGE AND RETRIEVAL SYSTEMS (Shree Choudhari¹, Sagar Wankhede² and Yash Prasad³)
- Nanotechnology-Enhanced Magnesium Alloys: An Updated Review (Shree Choudhari¹, Soham Mantri², Sagar Wankhede³)
- Characterization Challenges, Strategies and Standards for Thinwalled Structures (Shree Choudhari¹, Sagar Wankhede², Pralhad Pesode³)
- Additive Manufacturing (3D Printing) for Sustainability (Shree Choudhari¹, Yash Prasad² and Sagar Wankhede³)

CERTIFICATIONS

1. Learn Siemens S7-1200 PLC & HMI from Scratch using TIA PORTAL
2. Basics of Embedded Systems and Robotics (E-Yantra IIT BOM)
3. Robotics and ROS 2 - Learn by Doing! Manipulators
4. Ultimate Fusion 360 Course
5. Electronics in Biomedical Engineering: Theory & Repair